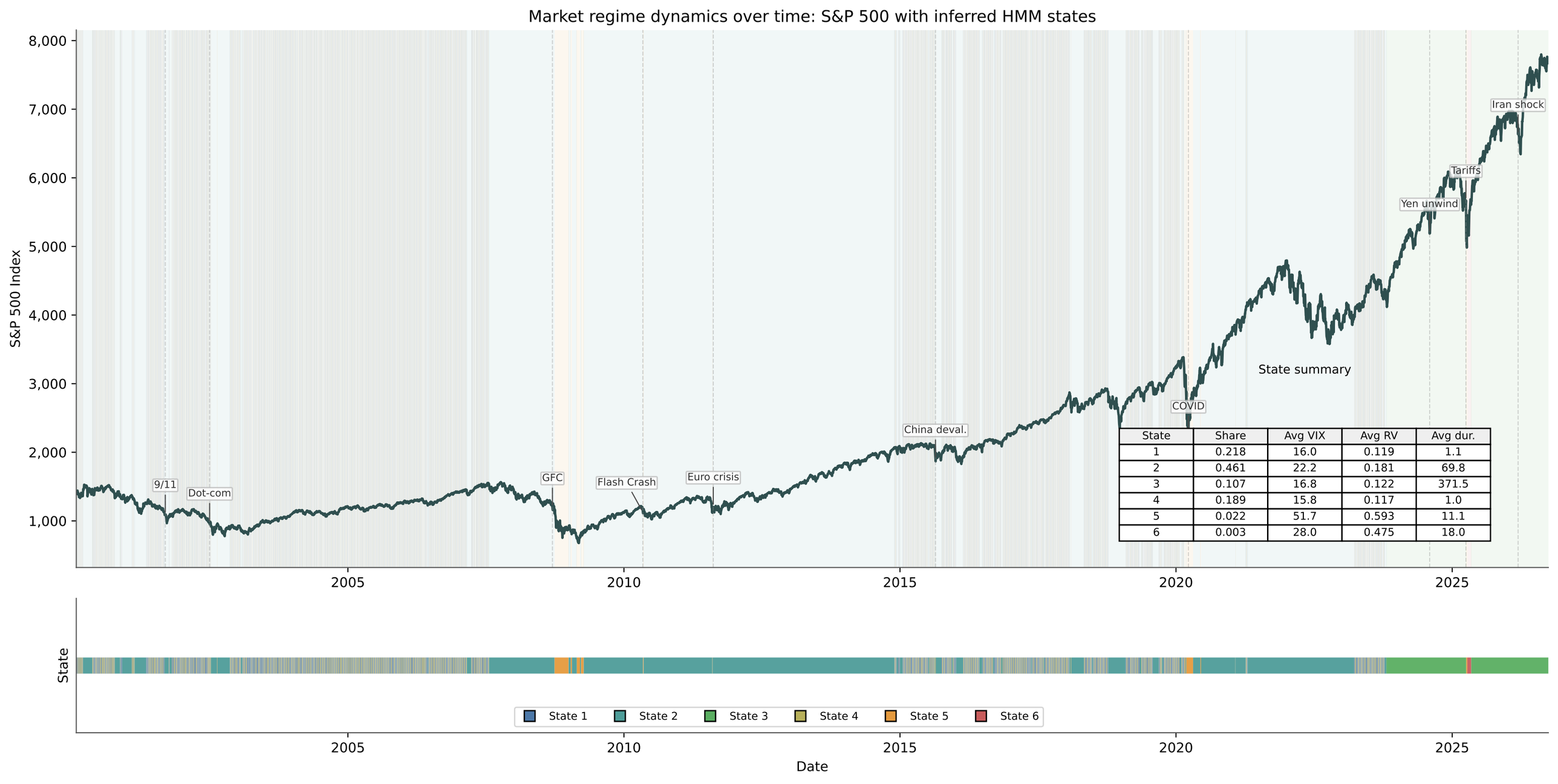

Market regime dynamics over time: S&P 500 with inferred HMM states
S&P 500 Index
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
9/11
Dot-com
GFC
Flash Crash
Euro crisis
China deval.
COVID
Yen unwind
Tariffs
Iran shock
State summary
State
Share
Avg VIX
Avg RV
Avg dur.
1 0.218 16.0 0.119 1.1
2 0.461 22.2 0.181 69.8
3 0.107 16.8 0.122 371.5
4 0.189 15.8 0.117 1.0
5 0.022 51.7 0.593 11.1
6 0.003 28.0 0.475 18.0
2005
2010
2015
2020
2025
State
State 1
State 2
State 3
State 4
State 5
State 6
Date